

Histogram Matching Numerical

Q. Perform histogram mapping in given image data with reference histogram.

Image data

2	1	1	2	2	1	1	2
2	7	7	0	0	0	2	3
2	6	6	7	0	1	4	4
1	6	5	5	5	4	4	4
1	5	4	4	4	5	5	4
1	5	5	5	6	7	4	3
2	4	5	6	7	7	7	6
2	4	4	5	5	6	6	5

Reference histogram

r	0	1	2	3	4	5	6	7
n	5	8	6	12	23	10	0	0

Step - 01: Calculate histogram of given image.

r	0	1	2	3	4	5	6	7
n	4	8	9	2	14	14	7	6

Step-2: Calculate the CDF of given image data. (Ha)

r	n	PDF	CDF
0	4	0.0625	0.0625
1	8	0.125	0.1875
2	9	0.140625	0.328125
3	2	0.03125	0.359375
4	14	0.21875	0.578125
5	14	0.21875	0.796875
6	7	0.109375	0.90625
7	6	0.09375	1

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Step-3: Calculate the CDF of reference histogram. (Hb)

r	n	PDF	CDF
0	5	0.078125	0.078125
1	8	0.125	0.203125
2	6	0.09375	0.296875
3	12	0.1875	0.484375
4	23	0.359375	0.84375
5	10	0.15625	1
6	0	0	1
7	0	0	1

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For matched image histogram.

r(Ha)	n(Ha)	s	n(s)
0	4	0	4
1	8	1	8
2	9	2	11
3	2	2	
4	14	3	14
5	14	4	21
6	7	4	
7	6	5	6
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Step-4: Matching table

r(Ha)	CDF(Ha)	CDF(Hb)	s
0	0.0625	0.078125	0
1	0.1875	0.203125	1
2	0.328125	0.296875	2
3	0.359375	0.484375	2
4	0.578125	0.84375	3
5	0.796875	1	4
6	0.90625	1	4
7	1	1	5

Step-5: Display image histogram after matching

r	0	1	2	3	4	5	6	7
n	0	0	0	45	0	127	74	66

Step-6: Replace image pixel values with matched gray level values.

2	1	1	2	2	1	1	2
2	5	5	0	0	0	2	2
2	4	4	5	0	1	3	3
1	4	4	4	4	3	3	3
1	4	3	3	3	4	4	3
1	4	4	4	4	5	3	2
2	3	4	4	5	5	5	4
2	3	3	4	4	4	4	4